

T.N.S.V.T

Documentacion Tecnica de Arquitectura

Symfony 8.1 | PHP 8.4 | Doctrine ORM | Mercure | Firebase FCM

Capacitor Android Web v8 | Capacitor Android Game v6 | MercadoPago | Binance Pay

Version: 1.0 | Fecha: 28/06/2026

Confidencial | Uso interno TNSVT

* . Indice General

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.
- 15.

1. Resumen Ejecutivo

Componentes Principales

Metricas Clave

Entidades:

Tablas:

Endpoints:

Servicios:

Comandos:

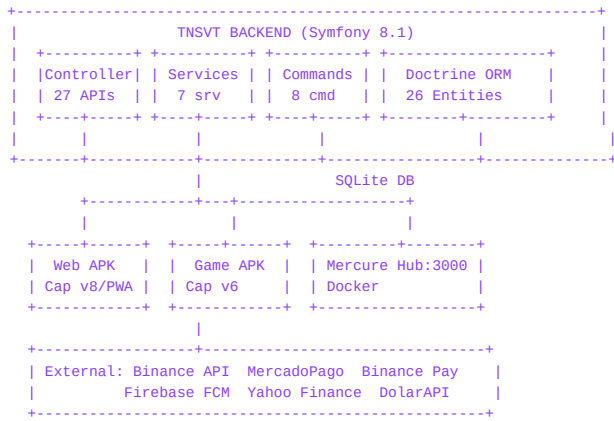
Frontend:

SW PWA:

Usuarios:

NOTA: Este documento cubre ambos APKs. Las secciones de pagos, autentificacion y notificaciones aplican al BACKEND COMPARTIDO, NO son exclusivas de ninguna aplicacion.

* 2. Arquitectura General



Flujo de Datos

* 3. Stack Tecnologico

Backend

Framework:
PHP:
ORM:
Database:
Security:
CORS:
Mailer:
Real-time:
HTTP:
Serialization:
Logging:
Templating:

Frontend Web (PWA)

HTML/CSS:
JavaScript:
API Client:
Charts:
PWA:
Biometrics:
Push Web:

Aplicaciones Moviles

Web APK:
Game APK:
JDK:
Build:

Servicios Externos

MercadoPago:
Binance Pay:
Binance API:
Firebase:
Mercure:
Yahoo Finance:
DolarAPI:

* 4. Estructura del Proyecto

```

tnsvt-symfony/
+-- assets/
|   +-- app.js           # Frontend principal (6,271 lines)
|   +-- api.js           # API client (326 lines)
|   +-- chart.js         # Chart TradingView + Mercure
|   +-- styles/app.css   # Estilos
+-- config/packages/     # Symfony bundles config
|   +-- routes.yaml      # Route loading (attribute-based)
+-- docker-compose.yml   # Mercure hub
+-- docs/                # PDF documentation
+-- migrations/          # 14 migration files
+-- public/
|   +-- sw.js            # Service Worker v37
|   +-- uploads/avatars/ # Avatar images
|   +-- downloads/       # APK files
|   +-- apk/             # APK versioned backups
+-- src/
|   +-- Command/         # 8 console commands
|   +-- Controller/Api/  # 27 API controllers
|   +-- Entity/          # 26 Doctrine entities
|   +-- Repository/      # Doctrine repositories
|   +-- Security/        # CodeAuthenticator, UserProvider
|   +-- Service/         # 7 business services
+-- templates/
|   +-- base.html.twig   # Single-page app template
|   +-- emails/          # Email templates (Twig)
+-- var/
|   +-- data_dev.db      # SQLite database (dev)
|   +-- log/             # Logs + debug emails
+-- composer.json        # PHP dependencies
+-- package.json         # npm dependencies
+-- .env                 # Environment config
+-- run-mercure.ps1      # Mercure start/stop script

```

* 5. Entidades del Sistema (26)

5.1 User (users)

```

    id            int (PK, auto)
    code          string(50), unique    # Codigo usuario (DEMO, ADMIN01)
    email         string(180), nullable
    name          string(100)
    active        boolean, default true
    lastLogin     DateTimeImmutable, nullable
    roles         json                  # ["ROLE_USER"]
    password      string(255), nullable # Solo admin
    walletBalance decimal(12,2)         # default 0.00
    diarySetupToken text, nullable      # Encryption key token
    diarySetupIv  string(48), nullable  # Encryption IV
    -----
    OneToMany WalletTransaction user.walletTransactions
    OneToMany TournamentEntry  user.tournamentEntries
    OneToMany DiaryEntry       user.diaryEntries
    OneToMany Connection       user.connections
    OneToOne  JournalSetting   user.journalSetting

```

5.2 Trade (trades)

Campos:

Relacion:

5.3 Tournament (tournaments)

Campos:

Relacion:

5.4 TournamentEntry (tournament_entries)

Campos:

Relacion:

5.5 WalletTransaction (wallet_transactions)

Campos:

Relacion:

5.6 Duel (duels)

Campos:

Relacion:

5.7 DuelRound (duel_rounds)

Campos:

Relacion:

5.8 FeedPost (feed_posts)

Campos:

Relacion:

5.9 Message (messages)

Campos:

Relacion:

5.10 Conversation (conversations)

Campos:

Relacion:

5.11 ConversationParticipant

Campos:

Relacion:

Entidades de Soporte (5.12 - 5.26)

5.12 AcademiaContent (academia_content)

Relacion:

5.13 AccessRequest (access_requests)

Relacion:

5.14 Connection (connections)

Relacion:

5.15 DiaryEntry (diary_entries)

Relacion:

5.16 Device (devices)

Relacion:

5.17 GameScore (game_scores)

Relacion:

5.18 JournalPermission (journal_permissions)

Relacion:

5.19 JournalSetting (journal_settings)

Relacion:

5.20 LikedPost (liked_posts)

Relacion:

5.21 MarketCandle (market_candle)

Relacion:

5.22 ModuleProgress (module_progress)

Relacion:

5.23 MonitorEvent (monitor_event)

Relacion:

5.24 Notification (notifications)

Relacion:

5.25 Task (tasks)

Relacion:

5.26 TraderProfile (trader_profiles)

Relacion:

* 6. Controladores API (27)

6.1 DuelController

6.2 SocialController

6.3 JournalController

6.4 TournamentController

6.5 MercadoPagoController

6.6 BinancePayController

Controladores Adicionales (6.7 - 6.27)

* 7. Servicios del Backend (7)

MercadoPagoService (162 lines)

[i] `file_get_contents` + `stream_context`. Notification URL desde `APP_SERVER_URL`.

BinancePayService (155 lines)

[i] Headers: `BinancePay-Timestamp`, `BinancePay-Nonce`, `BinancePay-Certificate-SN`.

PushNotificationService (234 lines)

[i] Prioriza v1 API con service account. Fallback legacy.

PushService (alternative)

[i] `titleForType()` mapea 10+ tipos a títulos en español.

TournamentMailer

[i] Twig en `templates/emails/`. Debug en `var/log/emails/`.

RealtimePublisher

[i] Usa Symfony Mercure Hub integration.

MercureSubscriberService

[i] JWT con `claim mercure.subscribe` y `SameSite=Strict`.

Comandos Console (8)

tournaments:process

app:seed-users

app:seed-trades

app:seed-tasks

app:seed-academia

tnsvt:reset-web-data

mercure:stream-candles

app:create-student

* 8. Frontend Web

8.1 assets/app.js (6,271 lines)

8.2 assets/api.js (326 lines)

8.3 templates/base.html.twig

* 9. Aplicaciones Móviles (APKs)

9.1 Web APK (com.tnsvt.app)

Framework:

Package:

Version:

Tamaño:

Build:

Sync:

Contenido:

APK file:

9.2 Game APK (com.tnsvt.market.instinct)

Framework:

Package:

Version:

Tamaño:

Build:

Contenido:

APK file:

Nota:

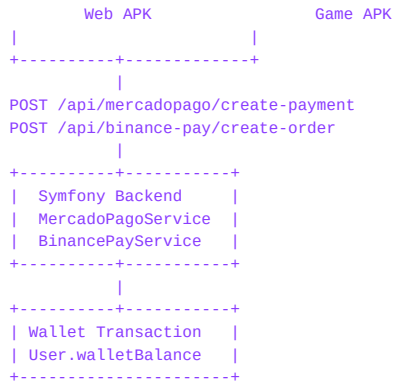
9.3 Backend Compartido

9.4 Service Worker (PWA)

* 10. Sistema de Pagos Compartido

IMPORTANTE: MercadoPago y Binance Pay son parte del BACKEND SYMFONY COMPARTIDO. NO son exclusivos del Game APK ni del Web APK. Ambos APKs pueden usar cualquiera de los dos metodos de pago.

10.1 Arquitectura de Pagos



10.2 MercadoPago Checkout Pro

Metodos MercadoPagoService

[i] MP_ACCESS_TOKEN debe configurarse en .env. Sin token, endpoints retornan 501.

10.3 Binance Pay (USDT)

Metodos BinancePayService

[i] BINANCE_PAY_API_KEY y BINANCE_PAY_SECRET_KEY en .env. Sin config, endpoints 501.

10.4 WalletTransaction - Tipos y Estados

```
TYPES:
TYPE_DEPOSIT      = "deposit"      # Credito manual/admin
TYPE_ENTRY_FEE    = "entry_fee"    # Fee de torneo
TYPE_PAYOUT       = "payout"       # Premio de torneo
TYPE_REFUND       = "refund"       # Reembolso
TYPE_WITHDRAW     = "withdraw"     # Retiro solicitado
TYPE_DUEL_ENTRY   = "duel_entry"   # Fee de duelo 1v1
TYPE_DUEL_WIN     = "duel_win"     # Premio de duelo
TYPE_DUEL_REFUND  = "duel_refund"  # Reembolso de duelo

STATUSES:
STATUS_PENDING    = "pending"      # Transaccion pendiente
STATUS_CONFIRMED  = "confirmed"    # Confirmada
STATUS_REJECTED   = "rejected"     # Rechazada
STATUS_REFUNDED   = "refunded"     # Reembolsada

METHODS:
METHOD_MANUAL_MP, METHOD_MANUAL_BINANCE, METHOD_MANUAL_CRYPT,
METHOD_AUTO_MP, METHOD_AUTO_BINANCE, METHOD_AUTO_CRYPT,
METHOD_GIFT, METHOD_OTHER
```

10.5 Admin Wallet (Credit/Debit)


```

+-----+ +-----+ +-----+
| MercureStreamCmd | ---> | RealtimePublisher | ---> | Mercure Hub |
| (PHP daemon 3s) |      | Symfony Service |      | Docker :3000 |
+-----+ +-----+ +-----+
                        |
                        +-----+
                        | SSE Stream |
                        | /chart/{ex}/{s} |
                        | /chart/ticker |
                        +-----+
                        |
                        +-----+
                        |
                        +-----+ +-----+
                        | Web APK PWA | | Game APK |
                        | chart.js | | HTML5 Canvas |
                        | EventSource | | EventSource |
                        +-----+ +-----+

```

MercureStreamCommand (mercure:stream-candles)

Metodo:

Auth:

MercureSubscriberService

```
getSubscribeUrl():
createSubscribeJwt():
createAuthCookie():
```

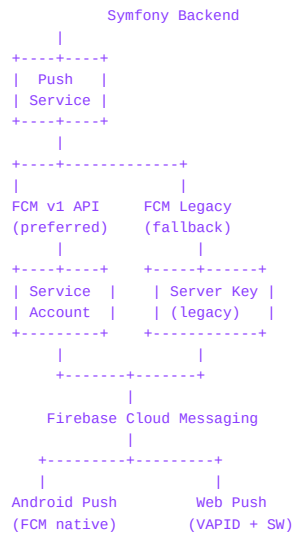
MercureController

11.4 Docker

TNSVT Trading Platform - Symphony 8.1 / PHP 8.4

* 12. Push Notifications (FCM)

12.1 Arquitectura Push



12.2 PushNotificationService (234 lines)

Modo v1 (preferido)

Modo Legacy (fallback)

12.3 PushService (alternativo)

12.4 Device Registration

Entity:

12.5 Notification Types

comment, like, post, mention	-> Feed social
signal	-> Senial de trading
dm	-> Mensaje directo
academia	-> Nuevo curso
task	-> Nueva tarea
access_request	-> Solicitud de acceso social
access_accepted	-> Solicitud aceptada
access_rejected	-> Solicitud rechazada
connection_removed	-> Conexion eliminada
permissions_changed	-> Permisos actualizados

* 13. Autenticacion y Seguridad

13.1 CodeAuthenticator

13.2 UserProvider

Metodo:

Metodo:

Metodo:

13.3 Admin Auth

13.4 Security Configuration

Password:

13.5 Journal Access Control

```
security:
  password_hashers:
    App\Entity\User: 'auto'
  providers:
    app_user_provider:
      id: App\Security\UserProvider
  firewalls:
    main:
      lazy: true
      provider: app_user_provider
      custom_authenticator: App\Security\CodeAuthenticator
      logout:
        path: /api/auth/logout
  access_control:
    - { path: ^/, roles: PUBLIC_ACCESS }
```

Visibilidad:

Permisos:

* 14. Base de Datos y Migraciones

14.1 Migraciones

```

2026-06-10 Version20260610213518.php # Usuarios iniciales + auth
2026-06-11 Version20260611003208.php # Wallet + transactions
2026-06-12 Version20260612144923.php # Torneos
2026-06-13 Version20260613051000.php # Feed + posts
2026-06-17 Version20260617171951.php # Chat + mensajes
2026-06-17 Version20260617192614.php # Notificaciones
2026-06-17 Version20260617193440.php # Academia + cursos
2026-06-17 Version20260617235932.php # Juego + scores
2026-06-22 Version20260622002308.php # MercadoPago + Binance
2026-06-22 Version20260622025850.php # Push notifications
2026-06-22 Version20260622025909.php # Music player
2026-06-25 Version20260625021044.php # Duelos 1v1 + rounds
2026-06-27 Version20260627140251.php # Diary + traits
2026-06-27 Version20260627214623.php # Journal social system

```

14.2 Diagrama de Relaciones (Simplificado)

```

users
|--- wallet_transactions (OneToMany)
|--- tournament_entries (OneToMany)
|--- diary_entries (OneToMany)
|--- connections (OneToMany)
|--- journal_settings (OneToOne)
|--- trader_profiles (OneToOne)
|--- devices (OneToMany)
|--- notifications (OneToMany)
|--- game_scores (OneToMany)
|--- module_progress (OneToMany)
|
tournaments
|--- tournament_entries (OneToMany)
|
duels
|--- duel_rounds (OneToMany)
|--- player1 -> User (ManyToOne)
|--- player2 -> User (ManyToOne)
|--- winner -> User (ManyToOne)
|
conversations
|--- messages (OneToMany)
|--- conversation_participants (OneToMany)
|
feed_posts
|--- liked_posts (OneToMany)
|--- author -> User (ManyToOne)

```

14.3 Composer Dependencies Clave

Doctrine:

Firestore:

CORS:

Mercure:

Mailer:

Security:

Serialization:

* 15. Roadmap y Proximos Pasos

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Completado] Junio 2026

[Pendiente] Corto plazo

[Pendiente] Corto plazo

[Pendiente] Mediano plazo

[Pendiente] Mediano plazo

[Pendiente] Mediano plazo

[Pendiente] Largo plazo

[Pendiente] Largo plazo

[Pendiente] Largo plazo

T.N.S.V.T

Documentacion Tecnica de Arquitectura

Fin del documento

Generado: 28/06/2026 | Symfony 8.1 / PHP 8.4 | TNSVT Platform